CS548T_06 Bayesian Decision Theory (BDT)

- Statistical approach to pattern classification.

⇒ to get optimal decision in problems (uncertainty).

The Goal of BDT:

Find optimal $g^*(x)$ for the given assumptions (loss function, priors, CCD):

↗ we have know that:

① prior: $p(y)$

② Class-conditional distribution $p(x|y)$

③ loss function $L(g(x), y)$

Framework:

a. World has states/classes, drawn as Y. $y \in Y$.

- prior: the probability of state y, $p(y)$.

b. observer measures observation/features from random variable X.

↗ we can only get the features X, not Y.

c. Class conditional distribution (CCD)

one CCD for each state:

$P(x|Y=y)$

d. decision function: use observations to make a decision about the state

$g(x) = X \to Y$

e. Loss function: penalizes for deciding wrong y (mistake)

$L(g(x), y) \geq 0$.

0-1 loss: $L(g(x), y) = \begin{cases} 0 & g(x)=y \\ 1 & g(x)\neq y \end{cases}$

Bayes Decision Rule:

- Risk - Expected value of the loss function

$$\text{Risk} = E_{XY}[L(g(x), y)] = \sum_y \int_x p(x, y) L(g(x), y)\, dx$$

$$= \int_x \sum_y p(x) p(y|x) L(g(x), y)\, dx$$

$$= \int_x p(x) \Big[\underbrace{\sum_y p(y|x) L(g(x), y)}_{\text{conditional Risk } R(x)\ (R(g(x)|x))\ \text{given observation } x}\Big] dx$$

$$= E_X[R(x)]$$

Since $p(x) \geq 0$, $L \geq 0$ then minimizing the risk can be achieved by minizing the conditional risk for each x, that is

For one x, $(\arg\min_{j \in Y} R(x))$

$$g^*(x) = g^* = \arg\min_{j \in Y} \sum_y p(y|x) L(j, y)$$

$$= \arg\min_{j \in Y} E_{Y|X}[L(j, y)]$$

↗ Bayes decision Rule. (BDR)

- Classification with 0-1 Loss:

$$L(g(x), y) = \begin{cases} 1 & g(x) \neq y \text{ (misclassification)} \\ 0 & g(x) = y \end{cases}$$

$Y = \{1, 2, 3, \cdots, C\}$

- Conditional Risk:

$$R(x) = E_{Y|X}[L(g(x), y)]$$

$$= \text{Probability of an error given } x$$

$$= Pr(g(x) \neq y | x)$$

$$= 1 - Pr(g(x) = y | x)$$

- BDR

$$g^*(x) = g^* = \arg\min_{j \in Y} R(x)$$

$$= \arg\min_{j \in Y} 1 - Pr(\overset{y=j}{g(x)=j} | x)$$

$$= \arg\max_{j \in Y} Pr(y=j|x)$$

↗ MAP rule: select the class with the largest probability of posterior.

⇒ By Bayes rule

- posterior $Pr(y=j|x) = \dfrac{P(x|y=j)P(y=j)}{P(x)}$

$$\Rightarrow g^*(x) = \arg\max_{j \in Y} Pr(y=j|x)$$

$$= \arg\max_{j \in Y} P(x|y=j) P(y=j)$$

$$= \arg\max_{j \in Y} \Big(\underbrace{\log P(x|y=j)}_{\text{CCD}} + \underbrace{\log P(y=j)}_{\text{prior}}\Big)$$

Example (2-classification (0,1))

$y \in \{0, 1\}$ Loss: 0-1 loss.

given x: pick 0 iff $P(x|0)P(0) > P(x|1)P(1)$

$$\Rightarrow \underbrace{\frac{P(x|0)}{P(x|1)}}_{\text{likelihood ratio}} > \underbrace{\frac{P(1)}{P(0)}}_{\text{threshold}} = T$$

(likelihood ratio test)

Summary above:

for 0-1 loss:

- BDR is the MAP rule
  (tells threshold for likelihood ratio test)
- Risk = Probability of error
- BDR minimizes probability of error.

This is called a generative classification model.

- use data to learn the CCDs
- use CCDs → decision rule.

Example 2: Noisy Channel

bit $y \in \{0,1\}$ → $\boxed{Tx}$ → $\boxed{\text{channel}}$

→ $\boxed{Rx}$ $x \in \mathbb{R}$ → decoder $y^* = \begin{cases} 0 & x < T \\ 1 & x \geq T \end{cases}$

Given X,

Model: $P(y=0) = P(y=1) = \frac{1}{2}$

CCD: Assume Gaussian additive noise

$X = \mu_y + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$

$p(x|y=0) = N(x | \mu_0, \sigma^2)$

$p(x|y=1) = N(x | \mu_1, \sigma^2)$

BDR, 0-1 Loss:

$$g^*(x) = \arg\max_{j \in Y} \log(p(x|y=j)) + \log p(y=j)$$

$$= \arg\min_j -2x\mu_j + \mu_j^2$$

Pick 0: $-2x\mu_0 + \mu_0^2 < -2x\mu_1 + \mu_1^2$

$$\Rightarrow x < \frac{\mu_1 + \mu_0}{2}$$

$$\boxed{T = \frac{\mu_1 + \mu_0}{2}}$$

- Intuitive threshold = place threshold halfway between means

- What if $p(Y)$ is not uniform?
  (prior is not uniform.),

  ⇒ No. $P(y=0) = P(y=1)$.
  we have $P(y=0) \neq P(y=1)$ ⇒

BDR: pick 0 if

$$\log p(x|0)\, p(0) > \log p(x|1)\, p(1)$$

$$\Rightarrow x < \frac{\mu_0 + \mu_1}{2} + \frac{\sigma^2}{\mu_1 - \mu_0} \log \frac{P(y=0)}{P(y=1)}$$

Gaussian Classification:

$$\begin{cases} Y = \{1, 2, \dots, C\} \\ X \in \mathbb{R}^d \\ P(y=j) = \pi_j \end{cases} \qquad p(x|y=j) = N(x | \mu_j, \Sigma_j)$$

BDR: $g(x) = \arg\max_j \log p(x|j) + \log p(j)$

$$= \arg\max_j \underbrace{-\frac{1}{2}\|x - \mu_j\|^2_{\Sigma_j} - \frac{1}{2}\log|\Sigma_j| + \log \pi_j}_{g_j(x)}$$

Special Case:

- Assume $\Sigma_j = \sigma^2 I$ (shared variance)

  define $g_j(x) = w_j^T x + b_j$

  where $\begin{cases} w_j = \frac{1}{\sigma^2}\mu_j \\ b_j = -\frac{1}{2\sigma^2}\mu_j^T \mu_j + \log \pi_j \end{cases}$

Geometric meaning:

Classes i, j shared a boundary if

$g_i(x) = g_j(x)$ ..

$(w_i - w_j)^T x + (b_i - b_j) = 0$ ⇒ hyperplane

vector (~~normal~~): $w = \frac{1}{\sigma^2}(\mu_i - \mu_j)$

$$x_0 = \left(\frac{\mu_i + \mu_j}{2}\right) + (\mu_j - \mu_i)\left[\frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \log \frac{\pi_i}{\pi_j}\right]$$

$w^T(x - x_0) = 0$